

SUMMARY OF ERGOGENIC AIDS AND RELATED SUBSTANCES*

Type of Supplement	Common Uses (Claims)	Typical Dosage**	Considerations Rating as Ergogenic Aid (1 to 5)***
5-hydroxytryptophan (5-HTP)	Used to increase serotonin levels in the brain to reduce appetite and carbohydrate craving; promote sleep; reduce depression	25 mg/day; studies used as much as 900 mg/day	Used instead of the amino acid tryptophan; potential association with fatal eosinophilia-myalgia syndrome needs clarification via additional research ¹ Rating: 3
Acetyl-L-carnitine (see also Carnitine)	Used to enhance brain function and reduce depression in elderly; antioxidant	1,000 mg/day	May benefit certain types of dementia more research needed ^{2,3} Rating: 3
Alpha-ketoglutarate	Used to prevent muscle protein breakdown during periods of stress such as recovery from surgery; anticatabolic	10–25 g/day	Can be converted to the amino acids glutamate and glutamine; some support for medical use as an anticatabolic; use as sports supplement needs more study; effective dosage likely impractical; similar to glutamine, may benefit athletes involved in heavy training ^{4,5} Rating: 1
Alpha-ketoisocaproate (KIC)	Precursor for the BCAA leucine; claims are made for prevention of muscle protein catabolism and reduction of fatigue during exercise	15–20 g/day	May have medical applications; unlikely to provide any ergogenic benefit beyond BCAA supplementation; primarily converted to leucine; effective dosage likely impractical ⁶ Rating: 1
Alpha-linolenic acid (also see DHA and EPA)	Increases omega-3 fatty acid intake; usual sources are flax oil, walnut oil, canola oil, and soybean oil	Not taken separately	Not equivalent to omega-3 fatty acids in fish oils; can be converted to the longer “fish oil” omega-3 fatty acids, but conversion may be too inefficient to attain optimal amounts for EPA and DHA ^{7,8} Rating: 1

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Androstenedione	Precursor for testosterone; intended to increase testosterone; enhance muscle protein synthesis	50–150 mg/day (used in research)	Claims not consistently supported by research; may ↑ estrogen in men more than testosterone; no evidence for ↑ protein synthesis; may produce positive urine test for nandrolone ^{9–11} Rating: 2
Arginine	↑ growth hormone; promotes muscle protein synthesis; enhances nitric oxide synthesis and relaxation of smooth muscle in blood vessels	1–10 g/day	Significance of ↑ growth hormone is questionable except for disease conditions; good potential for reducing the risk of heart attack and possibly stroke ^{12,13} Rating: 2.5
Aspartate salts (potassium magnesium aspartate or PMA)	Delay fatigue and increase endurance time to exhaustion	7–13 g of PMA divided into 2 or 3 doses	Large dose needed; research results mixed; benefit more likely in untrained people than in trained individuals ^{14,15} Rating: 2.5
Beta-alanine	Used to buffer acids in the body, which are increased with exercise	2–5 g/day	Serves as the precursor to carnosine, which buffers acid in the body; studies show increases in muscular endurance ^{16,17} and anaerobic running capacity ^{17,18} Rating: 4
Beta-carotene	Antioxidant; prevention of exercise-induced muscle soreness; enhanced recovery from exercise stress; precursor to Vitamin A	5–20 mg/day	Additional research is needed to understand possible health risks vs benefits of beta-carotene supplementation ^{19,20} ; safe Upper Level set by Expert Group on Vitamins & Minerals (UK) is 7 mg/day ²¹ Rating: 1
Branched-chain amino acids (BCAA) (leucine, isoleucine and valine)	Suppress muscle protein degradation; enhance exercise recovery; reduce perception of fatigue during endurance exercise	3–6 g/day at least half leucine; split pre- and post-exercise	Some research (but not all) indicates prevention of muscle protein breakdown as well as reduced fatigue by inhibiting tryptophan uptake into the brain ^{22–24} Rating: 3.5
Beta-hydroxy-beta-methylbutyrate (HMB)	Typically used to reduce protein degradation and promote muscle protein accretion during resistance training	3 g/day in combination with intense resistance exercise	Research results equivocal; may benefit untrained individuals more than trained; possibly effective for enhancing recovery of damaged muscle; apparently safe ^{25–29} Rating: 3

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Bromelain	Complex of proteolytic enzymes extracted from the base of pineapple plants; used to aid protein digestion or as an anti-inflammatory agent	Variable; dependent on specific preparation	Digestive aid function questionable; some of this enzyme escapes digestion and affects eicosanoids in a way that can reduce inflammation ^{30–34} Rating: 3
Caffeine	Stimulates central nervous system; improves reaction time, alertness, concentration; enhances mobilization of fat from fat cells; improves endurance performance	3–6 mg/kg body weight	Claims are supported by research; diuretic effect increases urine production at rest, but not during exercise; combination with ephedrine sources like ephedra (ma huang) may increase risk of adverse reactions ^{35–37} Rating: 4
Capsaicin	Used for weight loss and fat loss by raising metabolism and decreasing appetite	6–700 mg/day	The spicy molecule in hot peppers; capsaicin has reduced weight, abdominal fat, and appetite ^{18,38} Rating: 3–4
Carnitine (L-carnitine) (also see Acetyl-L-carnitine)	Enhances utilization of fat (fatty acids) as an energy source and improves endurance by decreased reliance on carbohydrate (muscle glycogen)	2–6 g/day	Some studies support claims, however others do not; potential benefit likely depends on dosage, length of time taking supplement, and type of performance being tested ^{39–43} Rating: 3
Casein	Principal protein of milk and primary protein component of cheese; used as a component of protein supplements	Depends on dietary protein intake and body weight	Casein has a lower biological value than whey protein, but casein has been shown to support more sustained protein synthesis due to more gradual digestion and absorption ^{44,45} Depends
Chitosan	A form of chitin extracted from the shells of crustaceans; used as a “fat blocker” to reduce absorption of dietary fat and cholesterol	1–4 g/day	Can help to reduce absorption of dietary cholesterol but apparently it does not bind enough fat to assist in weight loss ^{46–48} Rating: 2–3

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Chondroitin sulfate (see also Glucosamine)	Promote joint health and treat osteoarthritis; usually combined with glucosamine sulfate	400–500 mg/day, usually combined with glucosamine	Perhaps the most thoroughly studied nutraceutical, generally shown to enhance joint health ^{48–51} Rating: 3–4
Chromium picolinate	Enhances loss of body fat and maintenance of lean tissue; maintains chromium status and normal insulin sensitivity	35–200 mcg/day of elemental chromium	Body composition claims not supported by research; chromium is an essential trace element needed for normal insulin function; the picolinate form is somewhat controversial due to mutagenic effects at high doses in isolated cell and animal studies ⁵² Rating: 1–2
Chromium nicotinate (chromium + niacin) Chromium chloride	Used to enhance loss of body fat with maintenance of lean tissue; maintain chromium status and normal insulin sensitivity	35–200 mcg/day of elemental chromium	Body composition claims not supported by research; chromium is an essential trace element needed for normal insulin function; these forms of chromium may be preferable to the picolinate form as a source of chromium in a dietary supplement ⁵² Rating: 1–2
Chrysin	Used by men to reduce the production of estrogen and increase testosterone levels	125–250 mg	Limited human research does not support a testosterone-stimulating effect of chrysin when consumed at typical doses ⁵³ Rating: Unknown
Citrulline	Used to increase blood flow through nitric oxide production	3–8 g/day	Citrulline increases serum arginine levels, which increases nitric oxide and blood flow, which increases training volume and decreases muscle soreness. ^{54–58} Rating: 3–4
Citrus aurantium—bitter orange (source of synephrine)	Used as a thermogenic to increase resting energy expenditure and enhance fat loss; contains synephrine, octamine, and tyramine	300–975 mg/day (with synephrine content from 2–20 mg)	Can increase blood pressure, especially in combination with stimulants like caffeine; interacts with several drugs; safety and effectiveness for human use requires further study; concentration of active components can vary among products ^{59–57} Rating: Unknown

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Coenzyme Q-10 (ubiquinone)	Used to enhance energy metabolism and exercise performance; also used for various medical purposes	60–150 mg/day divided into 2–3 doses/day	Does not clearly affect exercise performance in normal healthy people; may benefit some disease conditions, but can have negative interactions with several drugs ^{58,59} Depends
Cola nut (see Caffeine)	Herbal source of caffeine		
Conjugated linoleic acid (CLA)	Used for fat loss and cancer prevention; most common food sources are milk and beef fat	1–8 g/day	This is a fatty acid that is likely beneficial to health; effectiveness in fat loss and cancer prevention requires further research ^{60,61} Rating: 2.5
Creatine	Used to enhance high-intensity exercise performance	2–25 g/day	Generally effective for enhancing maximal power/strength exercise performance; apparently safe for short term use; safety of long term-use and use by children and adolescents not known, beneficial for muscle mass gains ^{62,63} Rating: 4–5
DHA (docosahexaenoic acid) (also see Alpha-linolenic acid and EPA)	Long chain omega-3 fatty acid found in fish oils and some algae oils; used to reduce the risk of cardiovascular disease and age-related macular degeneration, protect brain function, and reduce plasma triglycerides	200–400 mg/day; typically taken combined with EPA in fish or algae oil	DHA is the major fatty acid in the brain and the retina; deficiency may adversely affect mental function and vision; supplementation combined with “blood thinner” drugs requires medical supervision; potential health benefits depend on a person’s essential fatty acid status; may increase vertical jump performance and decreases muscle soreness ^{7,8,64,65} Rating: 3
DHEA (dehydroepiandrosterone)	Used to increase levels of steroid hormones that may increase protein synthesis; slow changes associated with aging	50–200 mg	Unlikely to enhance muscle mass and strength gains; presently banned by WADA and USADA; potential benefits and risks require additional clarification; use only with medical supervision; some preparations can give positive drug test ^{65–68} Rating: 1–2

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EPA (eicosahexaenoic acid) (also see Alpha linolenic acid and DHA)	Long chain omega-3 fatty acid found in fish oils and some algae oils; used to reduce the risk of cardiovascular disease	200–500 mg/day; typically taken combined with DHA in fish or algae oil	EPA has important functions and can be elongated to DHA; supplementation combined with “blood thinner” drugs requires medical supervision; potential health benefits depend on a person’s essential fatty acid status ^{7,8} Rating: 3
Ephedra (ma huang)	Used for weight loss (usually combined with a source of caffeine), enhancing athletic performance, and treatment of allergies and asthma	8 mg ephedrine alkaloids taken up to 3 times a day	Causes slight increase resting metabolic rate; banned substance for sports competition; may be unsafe for some people, especially during exercise and when combined with caffeine; illegal in the United States for sale as dietary supplements ⁶⁹ Rating: 3–4
Ginkgo biloba (leaf extract standardized to 24% ginkgo flavanoid glycosides and 6% terpenes)	Major use is to enhance mental function in elderly who have limited blood circulation to the brain	40–80 mg of extract taken up to 3 times a day	Evidence for enhanced mental function is inconsistent; do not use during pregnancy, lactation; those on medication and those managing blood disorders should avoid use without medical supervision, however, this concern warrants further study ^{70–72} Depends
Glucosamine	Used to treat joint problems such as those associated with osteoarthritis and to prevent development of age-related joint problems; commonly taken in combination with Chondroitin	1,500 mg/day divided into 3 daily doses of 500 mg each	Generally considered safe and potentially effective for adjunctive treatment of osteoarthritis ⁷³ Rating: 3–4
Glutamine	Used for a variety of purposes, such as gastrointestinal support, prevention of muscle wasting, post-exercise recovery, immune system support	2–20 g/day	Nonessential amino acid produced by muscle protein catabolism; used by gastrointestinal tract and immune system as energy source; supplementation beneficial for some conditions ^{74,75} Rating: 2

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Green tea extract	Used for weight control and general health promotion; has replaced ephedra as key ingredient in many weight loss products	100–600 mg standardized to 50% catechins	Potentially beneficial for intended uses; can interact with many drugs; avoid use with medications without medical supervision ^{76,77} Rating: 3
HMB (hydroxymethylbutyrate or beta-hydroxy beta-methylbutyrate)	Used to reduce protein degradation and promote muscle protein accretion during resistance training	1–3 g/day	Likely beneficial and apparently safe; possibly effective for enhancing recovery of exercise-damaged muscle ^{28,29} Rating: 3–4
HCA (hydroxycitric acid) (<i>Garcinia cambogia</i> is a natural source of HCA)	Used for weight loss by inhibiting fatty acid synthesis, enhancing fat oxidation, and reducing appetite	250–1,500 mg HCA/day	Some research supports appetite control and enhanced fat oxidation effects; possibly enhances endurance; fat loss not supported by all studies ^{78–82} Rating: 2–3
Inositol	Lipotropic substance used to enhance liver handling of increased fatty acid levels during weight loss	1,000 mg	Human research is lacking to support the claims; may be beneficial in combination with other lipotropic substances; potential benefit to some mental disorders ^{83,84} Rating: 1–2
Iron	Essential for normal oxygen delivery to cells, energy production in cells, liver metabolism of toxins, etc.	Variable, depends on iron status	Iron supplementation can enhance exercise performance in individuals with anemic and nonanemic iron deficiency; medical assessment of iron status and supervision of iron supplementation is advisable ⁸⁵ Rating: 5 (if iron deficient)
Leucine (L-leucine) (see Branched-chain amino acids)			
Lipoic acid (alpha lipoic acid)	Used as an antioxidant and by people with diabetes to lower blood glucose and prevent complications	100–800 mg/day	Some research supports antioxidant claims; fairly high doses are likely required to benefit diabetes; may improve blood flow and provide anti-inflammatory benefits ^{86–89}

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Ma huang (see Ephedra)			
Medium-chain triglycerides (MCTs)	Used to facilitate weight loss and to enhance endurance performance	20–85 g/day	Increased metabolic rate via increased post-prandial energy expenditure ⁹⁰ and decreased fat mass ⁹¹ indicate weight loss benefits, though minor. MCTs could replace other calories, not just dietary fat, and this is more practical presently since MCT offerings are more abundant and available. Improvements in fat oxidation ⁹² could result in a glycogen-sparing effect (similar to carnitine) but more research would be needed. Rating: 2
Potassium magnesium aspartate (PMA) (see Aspartate salts)			
Pyruvate (pyruvic acid)	Used as a weight loss aid and to enhance endurance performance	7–50 g/day (doses used in research)	Claims are not consistently supported by research; very large amounts were used in studies showing benefit, which often caused intestinal distress ⁹³ Rating: 2–3
Sodium bicarbonate	Enhance performance of high- intensity exercise that is highly dependent on anaerobic glycolysis (lactic acid system)	300 mg/kg body weight taken 90 minutes prior to exercise	Potential for gastrointestinal distress such as nausea and/or diarrhea; correct dosing important for safety; familiarization with use of lower doses in training suggested ⁹⁴ Rating: 4.5
Sodium chloride	Enhances endurance exercise performance that lasts long enough to significantly deplete body sodium through sweat loss (generally longer than 4 or 5 hours)	Highly variable; depends on individual sweat rate and sodium loss in sweat	Sweat rates and sodium loss in sweat can vary greatly among athletes. Very heavy training can cause losses as much as 10,000 mg of sodium per day (~5 teaspoons of salt); muscle cramping can result from sodium depletion and those who tend to cramp typically have greater sodium losses in sweat ⁹⁵ Rating: 5

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Sodium citrate	Enhances performance of high- intensity exercise that is highly dependent on anaerobic glycolysis (lactic acid system); also, see Sodium bicarbonate	500 mg/kg body weight taken ~ 120 minutes prior to exercise	Potential for gastrointestinal distress such as nausea and/or diarrhea; correct dosing important for safety; familiarization with use of lower doses in training suggested; higher dose needed than with sodium bicarbonate ⁹⁶ Rating: 4
Taurine	Used for management of diabetes, heart problems, and miscellaneous other health problems	3–6 g/day	Nonessential amino acid that is synthesized in the body and is not essential in the diet; some evidence for benefits to heart problems and hypertension; limited research to support ergogenic claims in athletes ^{97,98,99} Rating: 2–3
Valine (see Branched-chain amino acids)			
Vanadyl sulfate	Used to enhance control of blood sugar level and enhance muscle development with strength training	Tolerable upper intake level for adults is 1.8 mg/day	Requirement for this mineral element is unknown; claims of benefit in healthy adults are not consistently supported by current research; effects on insulin sensitivity warrant further study ^{99,100} Rating: 1–2
Zinc Magnesium Aspartate (see Aspartate salts)	Used before sleep to increase anabolic hormones	Ingredient doses vary	The zinc and magnesium could be beneficial with individuals who are deficient; otherwise the studies are equivocal for increased hormones in healthy individuals. ^{101,112} Rating: 2

*The information in this table is for instructional purposes only to describe the major purported uses of these components of dietary supplements. Many of these substances are powerful chemicals and should be used only with medical guidance and proper dosage. Caution is especially important for anyone using medication, for children, and for women during pregnancy and lactation.

**Typical dosages are based on current popular use and/or doses used in research. Safety issues may exist for some of these doses even for nutrients. For many ergogenic aids, it is best to consult with a health-care professional for dosage concerns related to an individual's goals and specific health status.

***Ratings of potential for efficacy for some or all of the claims based on current research perspectives and a certain amount of unavoidable subjective judgment: 5 = Very Good, 4 = Good, 3 = Fair, 2 = Poor, 1 = Very Poor, ? = Not Known, Depends = Depends on conditions and purpose of use.

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APPENDIX D

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